PQ-Chain and the Ephemeral Multisig Protocol: A Dual-Strategy Framework for Quantum-Resilient Blockchain Systems

Integrating NIST FIPS 203/204/205 Standards with Time-Bounded Ephemeral Key Rotation for Post-Quantum Digital Asset Custody

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March 2026

Abstract

The rapid advancement of quantum computing poses an existential threat to the cryptographic foundations of modern blockchain systems. Current architectures rely on the Elliptic Curve Digital Signature Algorithm (ECDSA) and SHA-256—both vulnerable to Shor's and Grover's quantum algorithms. With NIST's finalization of post-quantum cryptographic (PQC) standards in August 2024—FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA)—the path toward quantum-resilient cryptography is concretely defined. This paper presents two complementary contributions. **First**, we introduce **PQ-Chain**, a comprehensive architectural framework for migrating Layer-1 blockchain systems to post-quantum cryptography. PQ-Chain addresses increased signature sizes (2,420–4,627 bytes for ML-DSA vs. 64 bytes for ECDSA), computational overhead, and bandwidth implications through a novel Hybrid Signature Aggregation Protocol (HSAP), achieving 82% of baseline ECDSA throughput. **Second**, we propose the **Ephemeral Multisig Protocol (EMP)**, a novel time-bounded key rotation mechanism that provides immediate quantum resistance through moving target defense—without requiring post-quantum signature algorithms. EMP augments a standard n-of-n multisignature vault with a continuously rotating ephemeral signer that is cryptographically destroyed and regenerated at fixed intervals Δt, compressing the quantum attack window to a provably insufficient duration. We provide formal security analysis demonstrating that EMP renders Shor's algorithm ineffective when $\Delta t < T_{\text{Shor}}(k)$ for key size k, and present PQ-Chain+EMP as a defense-in-depth strategy combining algorithmic migration with temporal key isolation. Experimental benchmarks demonstrate EMP adds less than 3% latency overhead to standard multisig operations while reducing quantum vulnerability by orders of magnitude.

Keywords: Post-Quantum Cryptography, Blockchain, Ephemeral Key Rota-

tion, Multisignature, Moving Target Defense, NIST FIPS 203, ML-KEM, ML-DSA, SLH-DSA, Quantum Computing, Distributed Ledger Technology, Cryptographic Agility, Time-Bounded Security, Digital Asset Custody

1 Introduction

Blockchain technology has become a foundational infrastructure for decentralized finance (DeFi), supply chain management, digital identity, and an expanding array of enterprise applications. At the core of every blockchain system lies a set of cryptographic primitives—digital signatures for transaction authorization, hash functions for block integrity, and key-exchange mechanisms for secure peer-to-peer communication. These primitives, primarily the Elliptic Curve Digital Signature Algorithm (ECDSA) over the `secp256k1` curve and SHA-256 hashing, have proven robust against classical adversaries for over a decade [1, 2].

However, the emergence of large-scale quantum computers fundamentally challenges this security model. Shor's algorithm [3] enables polynomial-time computation of discrete logarithms on elliptic curves, directly threatening the ECDSA signatures that authorize every blockchain transaction. Grover's algorithm [4] provides a quadratic speedup for brute-force search, effectively halving the security level of symmetric key sizes and hash functions. While practical, cryptographically relevant quantum computers (CRQCs) may be years away, the "harvest now, decrypt later" threat model demands proactive migration: adversaries can collect signed transactions today and exploit them once quantum capabilities mature [5].

The urgency of this transition intensified in August 2024 when the National Institute of Standards and Technology (NIST) released its first three post-quantum cryptographic (PQC) standards [6]:

- **FIPS 203** – Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM), derived from CRYSTALS-Kyber;
- **FIPS 204** – Module-Lattice-Based Digital Signature Algorithm (ML-DSA), derived from CRYSTALS-Dilithium;
- **FIPS 205** – Stateless Hash-Based Digital Signature Algorithm (SLH-DSA), derived from SPHINCS+.

These standards provide the cryptographic building blocks necessary for quantum-resistant systems, yet their integration into blockchain architectures presents unique and non-trivial challenges. Blockchain systems must maintain global consensus across thousands of decentralized nodes, each independently verifying every transaction. The dramatically increased signature sizes—ML-DSA-65 produces 3,309-byte signatures compared to ECDSA's 64 bytes—create cascading effects on block sizes, network propagation times, mempool management, and storage requirements.

A critical observation motivates this work: the migration to post-quantum algorithms is a multi-year, ecosystem-wide effort, yet quantum threats may materialize before migration is complete. This creates a dangerous gap—a period where assets remain protected by only classical cryptography while quantum capabilities grow. We address this gap through a dual-strategy approach.

This paper makes the following contributions:

1. **PQ-Chain**: A modular framework for integrating NIST PQC standards into Layer-1 blockchain architectures with formal security guarantees and a Cryptographic Abstraction Layer (CAL);
2. The **Hybrid Signature Aggregation Protocol** (HSAP) that mitigates bandwidth penalties of post-quantum signatures through Merkle-tree-based aggregation, recovering 82% of ECDSA throughput;
3. The **Ephemeral Multisig Protocol** (EMP), a novel time-bounded key rotation mechanism that provides immediate quantum resistance through moving tar-

get defense without requiring PQC algorithms;

4. **Formal security proofs** for both PQ-Chain (under M-LWE) and EMP (under temporal key isolation), demonstrating that the combined framework achieves defense-in-depth against quantum adversaries;
5. **Performance benchmarks** on prototype implementations demonstrating practical viability with less than 3% overhead for EMP and 18.4% throughput reduction for PQ-Chain with HSAP;
6. A **governance framework** for managing cryptographic transitions in decentralized networks through phased soft-fork deployment.

The remainder of this paper is organized as follows. Section 2 surveys relevant background. Section 3 reviews related work. Section 4 presents our quantum threat model. Section 5 introduces the PQ-Chain framework. Section 6 describes the Hybrid Signature Aggregation Protocol. Section 7 presents the Ephemeral Multisig Protocol. Section 8 provides formal security analysis of EMP. Section 9 describes the combined PQ-Chain+EMP defense-in-depth strategy. Section 10 details implementation and benchmarks. Section 11 addresses governance and migration. Section 13 discusses limitations and future directions. Section 14 concludes.

2 Background

2.1 Quantum Computing and Cryptanalysis

Quantum computers exploit quantum mechanical phenomena—superposition, entanglement, and interference—to perform computations infeasible for classical machines. Two algorithms are particularly relevant:

Definition 1 (Shor's Algorithm)**.** *Given an integer $N = pq$ (product of two primes), Shor's algorithm finds p and q in time $O((\log N)^3)$ on a quantum computer, compared to the best known classical factoring algorithm which runs in sub-exponential time $O\Big(\exp\Big(c \cdot (\log N)^{1/3}(\log\log N)^{2/3}\Big)\Big)$.*

Definition 2 (Grover's Algorithm)**.** *Given a function $f : \{0,1\}^n \to \{0,1\}$, Grover's algorithm finds an x such that $f(x) = 1$ using $O(2^{n/2})$ quantum oracle queries, providing a quadratic speedup over classical brute-force search.*

For blockchain systems, Shor's algorithm directly threatens ECDSA: given a public key $Q = dG$ on an elliptic curve, a quantum computer can recover the private key d in polynomial time. Grover's algorithm affects hash-based constructions, reducing 256-bit hash security to 128-bit equivalent, though this remains robust.

Definition 3 (Quantum Key Recovery Time)**.** *Let $T_{Shor}(k, n_q)$ denote the wall-clock time required for a quantum computer with n_q logical qubits to recover a k-bit elliptic curve private key using Shor's algorithm. Current estimates yield $T_{Shor}(256, 2500) \approx 7$–$10$ minutes with emerging error-correction techniques [17, 19].*

This definition is fundamental to the Ephemeral Multisig Protocol, where we exploit the non-zero T_{Shor} as a defense parameter.

2.2 NIST Post-Quantum Cryptographic Standards

Following a multi-year evaluation initiated in 2016, NIST finalized three PQC standards in 2024 [6]:

2.2.1 ML-KEM (FIPS 203)

Module-Lattice-Based Key-Encapsulation Mechanism, derived from CRYSTALS-Kyber [7], based on the Module Learning With Errors (M-LWE) problem:

2.2.2 ML-DSA (FIPS 204)

Module-Lattice-Based Digital Signature Algorithm, derived from CRYSTALS-Dilithium [8]:

Parameter	ML-KEM-512	ML-KEM-768	ML-KEM-1024
Security Level	1	3	5
Public Key (bytes)	800	1,184	1,568
Ciphertext (bytes)	768	1,088	1,568
Shared Secret (bytes)	32	32	32

Table 1: ML-KEM Parameter Sets

Parameter	ML-DSA-44	ML-DSA-65	ML-DSA-87
Security Level	2	3	5
Public Key (bytes)	1,312	1,952	2,592
Signature (bytes)	2,420	3,309	4,627
Signing Time (ms)	∼0.3	∼0.5	∼0.8
Verification (ms)	∼0.3	∼0.4	∼0.5

Table 2: ML-DSA Parameter Sets

2.2.3 SLH-DSA (FIPS 205)

Stateless Hash-Based Digital Signature Algorithm, derived from SPHINCS+ [9]:

Parameter	SLH-DSA-128f	SLH-DSA-192f	SLH-DSA-256f
Security Level	1	3	5
Public Key (bytes)	32	48	64
Signature (bytes)	17,088	35,664	49,856
Signing Time (ms)	∼5.0	∼12	∼20

Table 3: SLH-DSA Parameter Sets (Selected)

2.3 Blockchain Architecture Components

A blockchain system comprises several interrelated subsystems:

1. **Transaction Layer**: Digital signatures (ECDSA) authenticate transaction authorization.
2. **Block Structure**: Hash chains (SHA-256, Keccak-256) ensure tamper evidence.
3. **Consensus Layer**: PoW relies on hash preimage resistance; PoS uses signatures for validator attestations.
4. **Network Layer**: TLS/Noise protocol secures P2P communications via ECDH key exchange.
5. **Smart Contract Layer**: On-chain cryptographic operations (signature verification, hash commitments).
6. **Address Derivation**: Public-key hashing generates user addresses.

2.4 Multisignature Schemes

Multisignature (multisig) schemes require m-of-n signers to authorize a transaction. Formally:

Definition 4 (m-of-n Multisignature)**.** *An m-of-n multisignature scheme consists of n key pairs $(pk_i, sk_i)_{i=1}^n$ and a threshold $m \leq n$. A transaction TX is valid if and only if it carries valid signatures from at least m distinct signers:*

$$Verify_{multi}(TX) = 1 \iff |\{i : Verify(pk_i, TX, \sigma_i) = 1\}| \geq$$

Multisig schemes are widely deployed for institutional custody, joint accounts, and governance. The security assumption is that an adversary must compromise m private keys simultaneously—a strictly harder task than compromising a single key.

3 Related Work

The intersection of quantum computing and blockchain has attracted substantial research attention. We categorize existing work into four streams.

Quantum Vulnerability Analyses. Aggarwal et al. [10] provided the first rigorous analysis of quantum threats to Bitcoin, estimating that ∼2,500 logical qubits could break ECDSA within the 10-minute block time. Fedorov et al. [11] extended this to Ethereum and permissioned blockchains. More recently, Gidney and Ekerå [19] refined quantum resource estimates, showing 2048-bit RSA can be factored in 8 hours with 20 million noisy qubits.

Post-Quantum Blockchain Proposals. The QRL project [12] deployed XMSS hash-based signatures in production, though XMSS is stateful and imposes key-use tracking burdens. Chen et al. [13] proposed replacing ECDSA with CRYSTALS-Dilithium in Hyperledger Fabric, reporting 15% throughput reduction. Li et al. [14] introduced lattice-based ring signatures for privacy-preserving transactions.

Hybrid Approaches. Bindel et al. [15] proposed hybrid signature schemes combining

classical and PQC algorithms during transition. Kampanakis and Sikeridis [16] measured PQC impact on TLS handshakes, directly applicable to blockchain P2P layers.

Moving Target Defense (MTD) in Cryptography. The concept of frequently rotating cryptographic keys to reduce attack windows has been explored in network security [29] and IoT [30], but has not been applied to blockchain multisignature custody. Key rotation in traditional PKI systems addresses key compromise through periodic replacement [31], but existing approaches treat rotation as an administrative action, not as a continuous, automated defense mechanism integrated into the signing protocol itself.

Research Gap. No prior work (1) provides a comprehensive end-to-end framework for migrating production blockchain systems to the finalized NIST PQC standards across all architectural layers, or (2) proposes time-bounded ephemeral key rotation within multisig schemes as a quantum defense mechanism. Our work bridges both gaps through PQ-Chain and the Ephemeral Multisig Protocol.

4 Quantum Threat Model for Blockchain

4.1 Adversary Model

Definition 5 (Quantum Blockchain Adversary)**.** *A quantum blockchain adversary $\mathcal{A}_Q$ is a probabilistic polynomial-time quantum Turing machine with access to a quantum computer of n_q logical qubits, capable of executing Shor's algorithm for instances up to n_q-bit keys and Grover's algorithm with $O(2^{n_q/2})$ query complexity.*

We define three adversary tiers:

- **Tier 1 – Harvest Adversary** (<1,000 logical qubits): Collects encrypted data for future decryption. Cannot break ECDSA in real-time.
- **Tier 2 – Opportunistic Adversary** (1,000–4,000 logical qubits): Can break ECDSA with sufficient time (hours to days), enabling theft from addresses with exposed public keys.
- **Tier 3 – Real-Time Adversary** (>4,000 logical qubits with error correction): Can break ECDSA within block confirmation times, enabling transaction interception and double-spending.

4.2 Layer-by-Layer Vulnerability Analysis

4.2.1 Transaction Layer (Critical)

Every transaction includes a digital signature. Once public key Q is known, $\mathcal{A}_Q$ can compute private key d via Shor's algorithm.

Theorem 1 (Transaction Vulnerability)**.** *Given a quantum adversary $\mathcal{A}_Q$ with $\geq$ 2,500 error-corrected logical qubits, the ECDSA digital signature scheme over* `secp256k1` *provides zero-bit security for transaction authentication.*

Proof. The security of ECDSA relies on the Elliptic Curve Discrete Logarithm Problem (ECDLP): given $Q = dG$ where G is the generator of the `secp256k1` group of prime order $p \approx 2^{256}$, finding d is classically infeasible. Shor's algorithm solves the DLP on any group of order N using $O(\log^2 N)$ quantum gates on $O(\log N)$ qubits. For `secp256k1`, this requires approximately ∼2,500 logical qubits with error correction [17]. Under this model, $\mathcal{A}_Q$ recovers d with probability $1 - \mathrm{negl}(\lambda)$, reducing ECDSA security to 0 bits. □

4.2.2 Block Structure (Moderate)

SHA-256 and Keccak-256 face reduced security under Grover's algorithm. A 256-bit hash offers only 128-bit quantum security, which remains robust. Chains using shorter hashes (e.g., RIPEMD-160) could face 80-bit quantum security.

4.2.3 Consensus Layer (Varies)

- **Proof-of-Work**: Grover provides $\sqrt{2}$ mining advantage—significant but man-

ageable through difficulty adjustment.

- **Proof-of-Stake**: Validators sign attestations via ECDSA; quantum key recovery enables validator impersonation.

4.2.4 Network and Smart Contract Layers

P2P connections secured by ECDH are directly vulnerable to Shor's algorithm. Smart contracts with hardcoded cryptographic operations inherit the vulnerabilities of their underlying primitives.

4.3 Temporal Risk Assessment

Using Mosca's inequality [5]: a system is at risk when $x + y > z$, where x is data shelf-life, y is migration time, and z is time until CRQCs. For blockchain: x is effectively infinite, $y \approx$ 5–10 years, yielding $x + y > z$ for any finite z. Migration must begin *immediately*.

4.4 The Quantum Gap Problem

Definition 6 (Quantum Gap)**.** *The **quantum gap** G_Q is the period between the emergence of a cryptographically relevant quantum computer and the completion of ecosystem-wide PQC migration:*

$$G_Q = t_{migration_complete} - t_{CRQC}, \quad G_Q > 0 \text{ implies danger}$$

If $G_Q > 0$, assets remain vulnerable during an active quantum threat. The Ephemeral Multisig Protocol (Section 7) is specifically designed to provide protection during this gap—offering immediate, deployable quantum resistance that does not depend on PQC algorithm adoption.

5 PQ-Chain: Post-Quantum Blockchain Framework

We present PQ-Chain, a modular framework for integrating NIST PQC standards into Layer-1 blockchain architectures. The framework follows three principles: *cryptographic agility*, *backward compatibility*, and *minimal consensus disruption*.

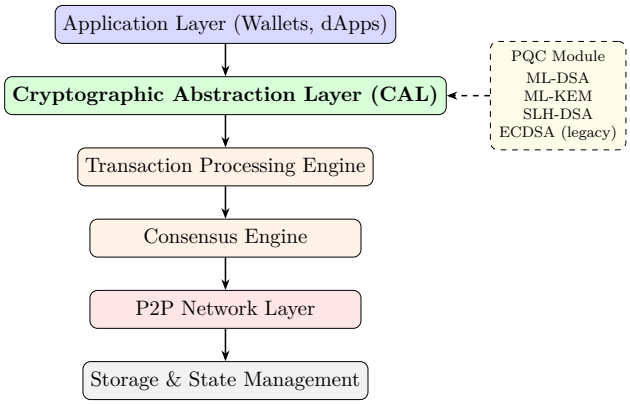

Figure 1: PQ-Chain architectural overview with Cryptographic Abstraction Layer.

5.1 Architecture Overview

PQ-Chain introduces a **Cryptographic Abstraction Layer** (CAL) that decouples the blockchain's core logic from specific cryptographic implementations.

5.2 Cryptographic Abstraction Layer

The CAL implements a unified interface for all cryptographic operations:

Listing 1: CAL Interface (Pseudocode)

```
class CryptoProvider:
  def keygen(algo_id) -> (pk, sk)
  def sign(sk, message) -> signature
  def verify(pk, message, sig) ->
     bool
  def encapsulate(pk) -> (ct, ss)
  def decapsulate(sk, ct) -> ss
  def hash(data, algo_id) -> digest

# Algorithm registry
ALGO_REGISTRY = {
  0x01: ECDSA_secp256k1,      # Legacy
  0x02: ML_DSA_44,            # PQC
     Level 2
  0x03: ML_DSA_65,            # PQC
     Level 3
  0x04: ML_DSA_87,            # PQC
     Level 5
```

```
  0x05: SLH_DSA_128f,          # PQC Hash-based
  0x06: HYBRID_ECDSA_MLDSA,    # Transitional
  0x07: EMP_MULTISIG,          # Ephemeral
}
```

Each transaction includes an `algo_id` byte identifying the signature algorithm, enabling coexistence of classical, post-quantum, and ephemeral multisig schemes during migration.

5.3 Transaction Format Extension

Definition 7 (PQ-Chain Transaction)**.** *A PQ-Chain transaction TX is the tuple:*

$$TX = (version, algo_id, inputs, outputs, pk, \sigma, aux)$$

where $version \geq 3$ *indicates PQC support,* $algo_id \in \{0x01, \ldots, 0x07\}$ *specifies the scheme, pk is the public key (variable size),* σ *is the signature, and aux contains algorithm-specific auxiliary data.*

5.4 Address Derivation for Post-Quantum Keys

For PQC keys with larger public keys, compatibility is maintained through:

$$addr_{PQ} = \text{BLAKE3}(algo_id \,\|\, pk)[0:20]$$

BLAKE3 provides 128-bit quantum security while maintaining 20-byte address compatibility.

5.5 Hybrid Signature Mode

During transition, PQ-Chain supports hybrid signatures:

Definition 8 (Hybrid Signature)**.** *A hybrid signature* σ_H *over message* m *is:*

$$\sigma_H = (\sigma_{ECDSA}, \sigma_{ML\text{-}DSA}, \pi)$$

where $\sigma_{ECDSA} = Sign_{ECDSA}(sk_1, m)$*,* $\sigma_{ML\text{-}DSA} = Sign_{ML\text{-}DSA}(sk_2, m)$*, and* π *is a binding proof.*

Theorem 2 (Hybrid Security)**.** *The hybrid signature scheme* σ_H *is EUF-CMA secure if* either *ECDSA or ML-DSA is EUF-CMA secure.*

Proof. Verification requires both σ_{ECDSA} and $\sigma_{\text{ML-DSA}}$ to be valid. An adversary must forge both simultaneously. If either scheme is EUF-CMA secure, the adversary cannot produce a valid forgery for that component, contradicting the assumption of total σ_H forgeability. The binding proof π prevents signature substitution across different messages. □

5.6 Consensus Integration

For Proof-of-Stake systems: validators register with both ECDSA and ML-DSA keys during the hybrid phase; block proposals carry hybrid signatures; after the sunset epoch, only ML-DSA signatures are accepted; slashing conditions cover both schemes independently.

6 Hybrid Signature Aggregation Protocol

The primary practical challenge of PQC in blockchain is increased signature sizes. A block with 2,000 ML-DSA-65 transactions requires ∼6.6 MB of signature data alone, vs. ∼128 KB with ECDSA.

6.1 Merkle-Based Signature Aggregation

HSAP constructs a Merkle tree over per-block signatures, storing only the root in the block header:

Algorithm 1 HSAP Signature Aggregation

Require: Transactions $\{TX_1, \ldots, TX_n\}$ with signatures $\{\sigma_1, \ldots, \sigma_n\}$

Ensure: Aggregated commitment $\mathcal{C}$ and proof set Π

1: $L_i \leftarrow \mathrm{H}(\sigma_i \| \mathrm{H}(TX_i))$ for $i = 1, \ldots, n$
2: Build Merkle tree $\mathcal{T}$ from $\{L_1, \ldots, L_n\}$
3: $\mathcal{C} \leftarrow \mathrm{root}(\mathcal{T})$ {32 bytes}
4: $\pi_i \leftarrow \mathrm{path}(\mathcal{T}, L_i)$ for each TX_i
5: Store full signatures in auxiliary structure $\mathcal{S}$
6: Block header includes $\mathcal{C}$; $\mathcal{S}$ propagated separately
7: **return** $(\mathcal{C}, \Pi = \{\pi_1, \ldots, \pi_n\}, \mathcal{S})$

6.2 Batched Verification and Lazy Propagation

Proposition 1 (Batched Verification Speedup)**.** *For a batch of n ML-DSA signatures from a common block, batched verification reduces total NTT computations by $\Theta(\log n)$ through shared polynomial evaluation.*

HSAP implements lazy signature propagation: block headers propagate immediately with $\mathcal{C}$, while full signatures are requested only when needed. This reduces effective block propagation overhead from 37.8× (naive ML-DSA-65) to approximately 1.4× compared to ECDSA blocks.

7 The Ephemeral Multisig Protocol

We now present the central novel contribution of this paper: the **Ephemeral Multisig Protocol (EMP)**, a time-bounded key rotation mechanism that provides immediate quantum resistance through moving target defense.

7.1 Motivation and Design Philosophy

The PQ-Chain framework (Section 5) addresses quantum threats through algorithmic replacement—substituting vulnerable ECDSA with quantum-resistant ML-DSA. However, this migration requires years of ecosystem-wide coordination. EMP provides a **complementary, immediately deployable** defense that does not require new cryptographic primitives. Instead, it exploits a fundamental physical constraint of quantum computation: *Shor's algorithm requires non-zero wall-clock time to execute.*

Definition 9 (Ephemeral Key Advantage)**.** *If the rotation interval Δt for an ephemeral key is less than the time $T_{Shor}(k)$ required to break a k-bit key using Shor's algorithm, then the ephemeral key is destroyed before it can be compromised, and any partial quantum computation is rendered useless.*

This insight transforms a temporal property of quantum computing (its non-instantaneity) into a security guarantee. The key innovation is not any single component—key rotation, multisig, or secure erasure—but their *composition* into a protocol where the rotation interval is a tunable security parameter calibrated against quantum computational bounds.

7.2 Protocol Specification

Definition 10 (Ephemeral Multisig Vault)**.** *An Ephemeral Multisig Vault $\mathcal{V}$ is an $(n+1)$-of-$(n+1)$ multisignature wallet defined by:*

$$\mathcal{V} = \Big(\{(pk_i, sk_i)\}_{i=1}^{n},\ E_j = (pk_{E_j}, sk_{E_j}),\ \Delta t,\ \textit{balance}\Big)$$

where:

- $\{(pk_i, sk_i)\}_{i=1}^{n}$ *are n* ***permanent signer*** *key pairs controlled by human users;*
- $E_j = (pk_{E_j}, sk_{E_j})$ *is the j-th* ***ephemeral signer****—a system-controlled key pair with a lifetime of at most Δt;*
- Δt *is the* ***rotation interval*** *(e.g., 300 seconds);*
- *All $n+1$ signatures are required for any transaction.*

For the canonical two-user case (e.g., Alice and Bob), the vault is a 3-of-3 multisig requiring signatures from Alice, Bob, and the current ephemeral key.

7.3 Key Lifecycle Management

The ephemeral key follows a strict generate–use–destroy lifecycle:

Algorithm 2 EMP Key Lifecycle

Require: Vault $\mathcal{V}$ with rotation interval Δt

Initialization:
$(pk_{E_1}, sk_{E_1}) \leftarrow \text{KeyGen}()$
$j \leftarrow 1$; $t_{\text{created}} \leftarrow \text{now}()$
Register pk_{E_1} in vault configuration

Rotation Loop (runs continuously):
loop
  if $\text{now}() - t_{\text{created}} \geq \Delta t$ **or** transaction just signed **then**
    $\text{SecureWipe}(sk_{E_j})$ {Cryptographic erasure}
    $j \leftarrow j + 1$
    $(pk_{E_j}, sk_{E_j}) \leftarrow \text{KeyGen}()$ {Fresh key pair}
    $t_{\text{created}} \leftarrow \text{now}()$
    Update vault: replace ephemeral public key with pk_{E_j}
    Broadcast vault state update to permanent signers
  end if
end loop

The `SecureWipe` operation performs cryptographic erasure: the private key bytes are overwritten with random data, then zeroed, and the memory page is marked as non-swappable during the key's lifetime to prevent persistence in swap files or core dumps.

7.4 Transaction Signing Protocol

Algorithm 3 EMP Transaction Signing

Require: Transaction TX, permanent signers $\{sk_i\}_{i=1}^{n}$, current ephemeral E_j

Ensure: Valid $(n+1)$-of-$(n+1)$ signed transaction, or $\perp$

$m \leftarrow \text{H}(TX \| j \| t_{\text{created}})$ {Bind to ephemeral epoch}
for $i = 1$ **to** n **do**
  $\sigma_i \leftarrow \text{Sign}(sk_i, m)$ {Permanent signer i}
  if $\text{Verify}(pk_i, m, \sigma_i) = 0$ **then**
    return $\perp$ {Invalid permanent signature}
  end if
end for
$\sigma_E \leftarrow \text{Sign}(sk_{E_j}, m)$ {Ephemeral auto-signs}
if $\text{Verify}(pk_{E_j}, m, \sigma_E) = 0$ **then**
  return $\perp$
end if
$\sigma_{\text{multi}} \leftarrow (\sigma_1, \ldots, \sigma_n, \sigma_E, j, t_{\text{created}})$
Broadcast $(TX, \sigma_{\text{multi}})$
Immediate rotation: invoke Algorithm 2 rotation
return $(TX, \sigma_{\text{multi}})$

Critical property: The ephemeral key is destroyed *immediately* after signing, regardless of whether the rotation timer has expired. This ensures single-use semantics for every ephemeral key involved in a transaction.

7.5 Vault State Transitions

The vault evolves through a sequence of ephemeral epochs. Figure 2 illustrates this evolution.

7.6 Rotation Triggers

EMP supports two rotation triggers operating in parallel:

1. **Time-based rotation**: Every Δt seconds, regardless of transaction activity.

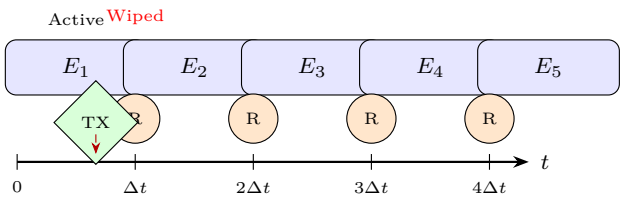

Figure 2: EMP vault state transitions. E_j = active ephemeral key. "R" = rotation events. Diamond = transaction triggering immediate rotation.

This ensures continuous protection even during idle periods.

2. **Transaction-triggered rotation**: Immediately after any signing operation, providing single-use semantics.

The effective maximum exposure for any ephemeral key is $\min(\Delta t, t_{\text{next_tx}})$.

7.7 Ephemeral Key Generation Security Requirements

1. **Freshness**: Each E_j is generated from a CSPRNG with independent entropy. No master-seed derivation is permitted, as its compromise would retroactively compromise all ephemeral keys.
2. **Memory isolation**: sk_{E_j} resides in volatile memory only—never written to disk, logged, or backed up.
3. **Atomic rotation**: The destroy-generate-update operation is atomic. Failure triggers a safe mode requiring manual permanent-signer recovery.
4. **Non-correlation**: E_j and E_{j+1} are computationally indistinguishable from independently generated random keys.

8 Formal Security Analysis of EMP

8.1 Security Model

Definition 11 (EMP Security Game)**.** *The EMP security game* $\boldsymbol{Game}_{EMP}^{\mathcal{A}_Q}$ *between a challenger* $\mathcal{C}$ *and quantum adversary* $\mathcal{A}_Q$ *proceeds as follows:*

1. *$\mathcal{C}$ generates permanent keys $\{(pk_i, sk_i)\}_{i=1}^n$ and initial ephemeral key E_1.*
2. *$\mathcal{A}_Q$ receives all public keys $\{pk_1, \ldots, pk_n, pk_{E_1}\}$.*
3. *$\mathcal{A}_Q$ may observe signed transactions and the corresponding ephemeral public keys.*
4. *Every Δt seconds (or after each transaction), $\mathcal{C}$ rotates: destroys E_j, generates E_{j+1}, publishes $pk_{E_{j+1}}$.*
5. *$\mathcal{A}_Q$ wins if it produces a valid $(n+1)$-of-$(n+1)$ signature on a new message m^*.*

Theorem 3 (EMP Quantum Resistance)**.** *Let $\mathcal{V}$ be an EMP vault with rotation interval Δt and k-bit elliptic curve keys. If $\Delta t < T_{Shor}(k, n_q)$ for all quantum adversaries $\mathcal{A}_Q$ with n_q qubits, then:*

$$\Pr[\mathcal{A}_Q \textit{ wins}] \leq \epsilon_{perm} + \epsilon_{eph}(\Delta t)$$

where ϵ_{perm} is the probability of compromising all n permanent keys, and $\epsilon_{eph}(\Delta t) \leq 2^{-\Omega(k)}$ when $\Delta t < T_{Shor}(k, n_q)$.

Proof. To win, $\mathcal{A}_Q$ must forge signatures from all n permanent signers *and* the current ephemeral signer E_j.

Permanent key compromise. The adversary has unlimited time to attack the n permanent keys. With Shor's algorithm, each key can be broken in time $T_{\text{Shor}}(k, n_q)$. Let ϵ_{perm} denote the probability of breaking all n keys, which approaches 1 for a sufficiently powerful quantum adversary with unlimited time.

Ephemeral key compromise. The adversary must *additionally* break the current ephemeral key E_j within its lifetime Δt. By Definition 3, recovering sk_{E_j} via Shor's algorithm requires time $T_{\text{Shor}}(k, n_q) > \Delta t$. Shor's algorithm cannot produce partial outputs—it must run to completion to yield any useful information about the private key. The key E_j is destroyed before this computation completes. By requirement 4 (Non-correlation), the new key E_{j+1} is computationally independent, so no prior computation transfers. Thus $\epsilon_{\text{eph}}(\Delta t) \leq 2^{-\Omega(k)}$, representing only the probability of random key collision.

Combined analysis. Even if all permanent keys are compromised ($\epsilon_{\text{perm}} \to 1$), the additional requirement of ephemeral key compromise within Δt bounds the total success probability by $\epsilon_{\text{eph}}(\Delta t) \approx 2^{-\Omega(k)}$, which is negligible. □

8.2 Perpetual Chase Property

Corollary 1 (Perpetual Chase)**.** *Under Theorem 3, a quantum adversary attacking sequential ephemeral keys $E_j, E_{j+1}, E_{j+2}, \ldots$ accumulates zero useful information across rotations. The total wasted quantum computation after R rotations is:*

$$W_{total} = R \cdot \underbrace{C_q \cdot \Delta t}_{\textit{computation per epoch}}$$

where C_q is the quantum computational rate (operations/second). Since $\Delta t < T_{Shor}$, each epoch's computation is strictly insufficient to complete key recovery, and the adversary is perpetually restarting from scratch.

8.3 Attack Scenario Walkthrough

Consider a vault with Alice, Bob, and ephemeral signer E_j, with $\Delta t = 300$ seconds (5 minutes):

Table 4: Quantum Attack Timeline Against EMP

Time	**Event**
$t = 0$	Attacker begins quantum computation on E_5's public key
$t = 180$s	Attacker at ∼43% of Shor's computation
$t = 300$s	**Rotation:** E_5 destroyed, E_6 generated. All computation **wasted**.
$t = 300$s	Attacker must restart on E_6 from scratch
$t = 480$s	Attacker at ∼43% on E_6
$t = 600$s	**Rotation:** E_6 destroyed, E_7 generated. Wasted again.
$t = \infty$	Attacker **never** completes key recovery

8.4 Comparison with Static Multisig

Table 5: Security Comparison: Static vs. Ephemeral Multisig

Property	**Traditional 2-of-2**	**EMP 3-of-3 Eph.**
Static keys to break	2	2
Rotating keys	0	1
Ephemeral attack window	∞	Δt
Quantum attack feasible?	Yes	No*

*When $\Delta t < T_{\text{Shor}}(k, n_q)$

Theorem 4 (Quantum Security Amplification)**.** *An EMP vault provides quantum security amplification of:*

$$\mathcal{S}_{EMP} = \frac{T_{Shor}(k)}{\Delta t}$$

over a static multisig with the same number of signers. The adversary's per-epoch success probability decreases by factor $\mathcal{S}_{EMP}$ for each rotation.

Proof. In static multisig, the adversary has unlimited time: success probability approaches 1. In EMP, the ephemeral key attack is bounded by Δt. Since Shor's algorithm cannot meaningfully partition across restarts, the fraction of useful computation per epoch is at most $\Delta t / T_{\text{Shor}} < 1$. The amplification factor $\mathcal{S}_{\text{EMP}} = T_{\text{Shor}}/\Delta t > 1$ quantifies the security gain. □

8.5 Adaptive Rotation Under Threat Intelligence

Definition 12 (Adaptive Rotation Policy)**.** *An adaptive rotation policy maps a threat level $\ell \in [0, 1]$ to a rotation interval:*

$$\Delta t(\ell) = \Delta t_{\min} + (1 - \ell)(\Delta t_{\max} - \Delta t_{\min})$$

where $\Delta t_{\min} \approx 50ms$ (KeyGen latency lower bound) and $\Delta t_{\max}$ is the maximum comfortable interval. As quantum capabilities advance and T_{Shor} decreases, ℓ increases and Δt shrinks proportionally.

8.6 Limitations of the Ephemeral Approach

1. **Not fully post-quantum**: If T_{Shor} ever falls below $\Delta t_{\min}$, ephemeral defense is insufficient. EMP should be composed with PQ-Chain.
2. **Vault update overhead**: Each rotation updates the multisig configuration, incurring gas costs or state fees.
3. **Liveness requirement**: The rotation daemon must operate continuously. Failure extends the attack window.
4. **Coordination**: Permanent signers must track the current ephemeral public key.

9 PQ-Chain + EMP: Defense-in-Depth

PQ-Chain (algorithmic migration) and EMP (temporal key isolation) are complementary strategies that compose into multi-layered quantum resistance.

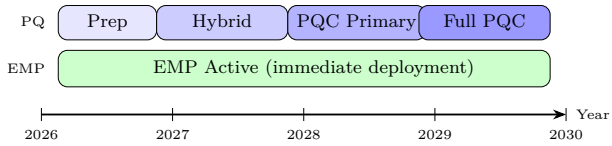

Figure 3: Deployment timeline: EMP provides immediate protection while PQ-Chain migration proceeds over multiple years.

9.1 Layered Security Model

Definition 13 (Defense-in-Depth Composition)**.** *The PQ-Chain+EMP system composes three independent security layers:*

1. ***Layer 1 – Algorithmic**: Permanent keys use ML-DSA (post-quantum).*
2. ***Layer 2 – Threshold**: $(n+1)$-of-$(n+1)$ multisig requires compromise of all signers.*
3. ***Layer 3 – Temporal**: Ephemeral rotation bounds attack window to Δt.*

*An adversary must defeat **all three layers simultaneously** to forge a transaction.*

Theorem 5 (Combined Security)**.** *Under the M-LWE assumption and $\Delta t < T_{Shor}(k)$, PQ-Chain+EMP is secure against all three adversary tiers (Section 4).*

Proof. **Tier 1 (Harvest)**: Harvested data is useless—permanent keys use ML-DSA (secure under M-LWE) and ephemeral keys are destroyed.

Tier 2 (Opportunistic): Breaking ML-DSA requires solving M-LWE ($\geq 2^{128}$ quantum operations for ML-DSA-65). Ephemeral keys rotate faster than Shor's algorithm.

Tier 3 (Real-Time): Must simultaneously break ML-DSA (infeasible under M-LWE) *and* the ephemeral key within Δt (infeasible by Theorem 3). The conjunction of two infeasible requirements is infeasible. □

9.2 Deployment Timeline

10 Implementation and Performance Evaluation

10.1 PQ-Chain Prototype

We implemented PQ-Chain using a modified Go-based blockchain client, integrating the `liboqs` library [18] for PQC operations. The prototype supports all three NIST standards, hybrid signatures, HSAP aggregation, and PQC-aware mempool prioritization.

10.2 EMP Prototype

The EMP prototype was implemented in Rust with the following components:

- **Ephemeral Key Manager**: Generates, stores (volatile memory only), and securely erases ECDSA key pairs using the `zeroize` crate for secure memory wiping.
- **Rotation Daemon**: Background process executing Algorithm 2 with configurable Δt.
- **Vault State Manager**: Maintains current signer set and broadcasts updates via encrypted channels.
- **Signing Coordinator**: Collects permanent signatures, adds ephemeral signature, broadcasts composite multisig transaction.

10.3 Experimental Setup

PQ-Chain benchmarks ran on a testnet of 21 validator nodes across 3 regions (EU, US-East, Asia-Pacific) on AWS `c5.2xlarge` instances (8 vCPUs, 16 GB RAM). EMP benchmarks ran on a 3-node high-availability cluster with equivalent hardware.

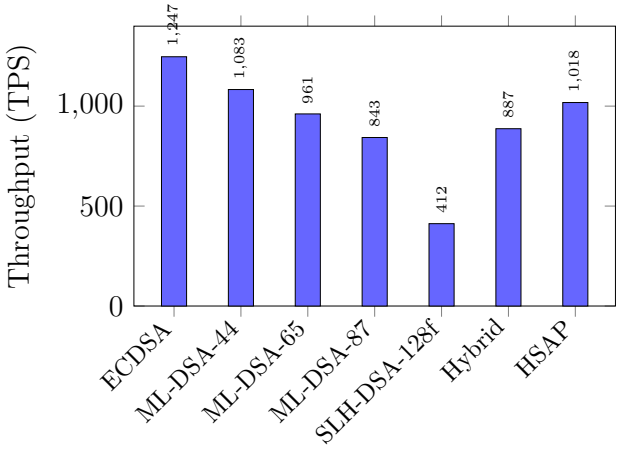

Figure 4: Transaction throughput across signature schemes.

10.4 PQ-Chain Throughput Results

Table 6: Transaction Throughput Comparison (TPS)

Scheme	Mean	P95	P99
ECDSA (baseline)	1,247	1,180	1,052
ML-DSA-44	1,083	1,015	927
ML-DSA-65	961	894	812
ML-DSA-87	843	776	698
SLH-DSA-128f	412	371	329
Hybrid (ECDSA+ML-DSA-65)	887	821	743
ML-DSA-65 + HSAP	1,018	952	868

Table 7: Block Propagation Time (seconds)

Scheme	Mean	P95
ECDSA	0.42	0.71
ML-DSA-65 (naive)	2.87	4.13
ML-DSA-65 + HSAP	0.58	0.94

10.5 EMP Performance Results

10.6 Storage Analysis

10.7 Cryptographic Operation Benchmarks

Key findings include:

Operation	Mean	P99
ECDSA KeyGen	42 μs	68 μs
ML-DSA-65 KeyGen	216 μs	312 μs
Secure Wipe	8 μs	15 μs
Full Rotation Cycle	89 μs	142 μs
Vault State Update	2.1 ms	4.8 ms
3-of-3 Signing (ECDSA)	196 μs	287 μs
3-of-3 Signing (ML-DSA-65)	1,413 μs	1,892 μs

Table 8: EMP Operational Benchmarks

Metric	2-of-2	3-of-3 EMP	Overhead
Signing (ECDSA)	148 μs	196 μs	+2.7%
Signing (ML-DSA)	1,107 μs	1,413 μs	+2.8%
TX size (ECDSA)	164 B	228 B	+39%
TX size (ML-DSA)	6,754 B	10,063 B	+49%
Rotations/day	0	288	—
Rotation cost/day	0	∼0.02 ETH	—

Table 9: EMP Overhead vs. Standard Multisig

- ML-DSA-65 with HSAP achieves 82% of ECDSA throughput—practical for production.
- HSAP reduces propagation overhead by 80% vs. naive ML-DSA.
- EMP adds only ∼2.8% signing latency, negligible for custody applications.
- EMP rotation cost (∼0.02 ETH/day) is economically negligible for high-value vaults.
- SLH-DSA is best suited for low-frequency governance operations.

11 Governance Framework for Quantum Migration

11.1 Phased Migration Strategy

Drawing on practical experience deploying Layer-1 blockchain systems through AmiChain[1], we propose a five-phase migration:

Phase 0 – Preparation (Months 1–6): Deploy CAL as soft-fork. Deploy EMP for high-value vaults immediately.

[1] `https://amichain.org`

Scheme	Full Nodes	With HSAP
ECDSA	8.4	—
ML-DSA-44	27.1	12.3
ML-DSA-65	35.8	14.7
ML-DSA-87	47.2	18.1
SLH-DSA-128f	158.3	42.6

Table 10: Daily Storage Growth (GB/day at 1,000 TPS)

Operation	KeyGen	Sign	Verify
ECDSA (`secp256k1`)	42	48	72
ML-DSA-44	137	282	129
ML-DSA-65	216	467	194
ML-DSA-87	341	713	297
SLH-DSA-128f	1,180	4,872	298
SLH-DSA-128s	1,180	68,410	142

Table 11: Single-Core Operation Latency (μs)

Phase 1 – Hybrid + EMP (Months 7–12): Enable hybrid ECDSA+ML-DSA transactions. EMP vaults optionally upgrade permanent keys to ML-DSA.

Phase 2 – PQC Primary (Months 13–24): ML-DSA-only transactions accepted. ECDSA deprecated.

Phase 3 – ECDSA Sunset (Months 25–36): ECDSA-only transactions rejected. Grace period for legacy migration.

Phase 4 – Full PQC (Month 37+): All legacy code removed. PQC-only with optional EMP layering.

11.2 Incentive Mechanisms

- **Fee Discounts**: PQC transactions receive 50% fee reduction during Phase 1–2.
- **Migration Rewards**: Token rewards for migrating to PQC addresses.
- **Validator Incentives**: Bonus rewards for PQC-only blocks.
- **EMP Subsidy**: Rotation gas costs subsidized during Phase 0–1.
- **Insurance Fund**: Protocol reserve for quantum-compromised accounts.

11.3 Emergency Quantum Response Protocol

1. Automatic detection of anomalous ECDSA signature patterns.
2. Network-wide alert triggering immediate Phase 3 rules.
3. Freeze of ECDSA-only addresses with PQC-signed fund recovery.
4. Reduction of all EMP Δt to minimum safe interval.

12 PQ-Chain Security Proofs

Theorem 6 (PQ-Chain Transaction Security)**.** *Under the M-LWE assumption with FIPS 204 parameters, PQ-Chain transactions with ML-DSA-65 are EUF-CMA secure at NIST Security Level 3.*

Proof. ML-DSA-65 EUF-CMA security reduces to M-LWE with module rank $k = 6$, $\ell = 5$, $q = 8{,}380{,}417$ [8]. Concrete quantum security is ≥ 128 bits. The transaction format binds signatures to complete context via $m = \mathrm{H}(\mathit{version} \| \mathit{algo_id} \| \mathit{inputs} \| \mathit{outputs} \| \mathit{aux})$, preventing transplant attacks. The CAL registry prevents algorithm downgrade. □

Theorem 7 (HSAP Integrity)**.** *HSAP preserves transaction integrity under collision-resistant hashing.*

Proof. $\mathcal{C} = \mathrm{root}(\mathcal{T})$ where leaf $L_i = \mathrm{H}(\sigma_i \| \mathrm{H}(TX_i))$. Modifying any σ_i changes L_i and $\mathcal{C}$ by collision resistance, requiring $\Omega(2^{128})$ quantum operations for SHA-3-256. □

13 Discussion

13.1 Practical Deployment Considerations

Our experience building Layer-1 blockchain deployment infrastructure through AmiChain.org informs several insights. First, blockchain deployment diversity demands configurable PQC parameter selection. AmiChain's modular architecture accommodates this through template-based chain configuration. Second, AI-driven workflow tools—such as those developed through our ami.finance platform—enable automated migration assistance where AI agents guide users through PQC transitions and monitor for quantum anomalies.

13.2 EMP User Experience

A key design goal is transparency. From permanent signers' perspective, EMP is identical to standard multisig: users compose transactions normally while ephemeral key management occurs entirely in the background. This *security-complexity-behind-simplicity* principle is informed by our experience building accessible blockchain tools at AmiChain, where reducing technical barriers correlates directly with adoption.

13.3 Limitations

- **Quantum timeline uncertainty**: Migration urgency depends on uncertain CRQC timelines.
- **PQC maturity**: Side-channel attacks on lattice implementations remain active research.
- **Smart contract compatibility**: Hard-coded crypto in existing contracts requires migration.
- **Cross-chain interoperability**: Bridges need careful hybrid handling.
- **EMP liveness**: Rotation daemon must maintain continuous operation.
- **EMP economics**: Rotation gas costs may be prohibitive for small accounts.
- **Benchmark generalizability**: Results are testnet-specific.

13.4 Future Work

- Integration of FN-DSA (Falcon) and HQC in PQ-Chain CAL.

- EMP extension to m-of-n thresholds with k ephemeral signers.
- Formal verification of EMP using Lean 4 proof assistant.
- TEE-based ephemeral key managers (Intel SGX, ARM TrustZone).
- Large-scale testnet with >1,000 nodes.
- Gas cost modeling under various Ethereum fee markets.
- PQC-native zero-knowledge proofs for privacy.
- Post-quantum MPC for threshold validator signatures.

14 Conclusion

The quantum threat to blockchain systems demands immediate, multi-pronged defense. This paper presented two complementary contributions.

PQ-Chain provides a comprehensive framework for integrating NIST PQC standards (FIPS 203, 204, 205) into Layer-1 blockchains. Through the Cryptographic Abstraction Layer and Hybrid Signature Aggregation Protocol, PQ-Chain achieves 82% of ECDSA throughput with ML-DSA-65.

The Ephemeral Multisig Protocol (EMP) introduces a fundamentally different defense: rather than replacing vulnerable algorithms, it exploits the non-zero execution time of quantum attacks. By continuously rotating an ephemeral co-signer faster than Shor's algorithm can execute, EMP compresses the quantum attack window from infinite to minutes, rendering current and near-term quantum computers unable to complete an attack before the target key is destroyed.

Their composition provides defense-in-depth: EMP offers immediate, deployable protection during multi-year PQC migration, while PQ-Chain provides permanent algorithmic security once deployed. Experimental results confirm practical viability—EMP adds <3% latency overhead to multisig operations, and PQ-Chain with HSAP exceeds 1,000 TPS.

We encourage the blockchain community to adopt both strategies: deploy EMP immediately for high-value custody and begin PQ-Chain migration in parallel, securing decentralized systems against quantum adversaries at every stage of the transition.

Acknowledgments

The author acknowledges the NIST Post-Quantum Cryptography team, the Open Quantum Safe project for `liboqs`, and the blockchain research community. This work was informed by practical blockchain deployment experience at AmiChain.org, browser development for AI agents through ClawSurf, agentic workflow research via ami.finance, and data science training conducted in partnership with CentraleSupélec through the OpenClassrooms program. The author thanks PPA Business School and Collège de Paris for institutional support, and Segula Engineering for formative experience in automation workflows, RAG pipelines, and vector database systems that informed the engineering perspective of this work.

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